Demand Control Project: Electrician One-Sheet

The Recording Club · 1660 9th Street, Santa Monica, CA 90404 · recording studio + members' club
Site contact: Taji (facilities, on-site lead) · taji@therecording.club · Greg Spero (owner) approves the written estimate; nothing is approved until it is signed off.

HARD RULE: Shed at 18.5 kW; never above 19.2 kW in any 15 min (meter 255000-006336).
Billing steps: SCE bills demand in 1.6 kW steps (19.2 kW bills as 19; 20.8 bills as 21), so the **controller setpoint must be about 18.5 kW** and the **true 15-min average 19.2 kW or less**.
Battery backstop: a battery is needed only for the controlled meter. Its interval-data peak was 27.2 kW, so a battery would have to cover about **8 kW** (27.2 down to about 19) to keep every 15-minute average under the limit.

Context: the main peak drivers are believed to be the 7 rooftop ACs (~22 kW total) stacking, and the sauna.

LOAD INVENTORY & SHED PRIORITY (METER 255000-006336)

Priority	Load	Breaker (IES)	Size
1st (Greg)	Sauna heater	30 A, 2-pole	6.2 kW (IES) vs 12.3 kW (TRC notes): confirm from nameplate
2nd	EV charger: Tesla Wall Connector, 240 V 1-ph; set to 11 A (~2.6 kW) for Greg's car; may draw more for other cars (rare).	TBD on site	~2.6 kW at 11 A
3rd stagger	AC #7 — **flagged as spiking**; model TBD	30 A, 2-pole	2.5–4.5 kW (est.)
3rd	AC #3: Carrier 48ESNA4809030, 4-ton, 220 V	30 A, 2-pole	4.5–5.0 kW (est.)
3rd	AC #2 (Mixing Room), 3–4 ton (est.); model TBD	30 A, 2-pole	3.5–4.5 kW (est.)
3rd	AC #6; model TBD	30 A, 3-pole	2.5–4.5 kW (est.)
3rd	AC #5; model TBD	40 A	2.5–4.5 kW (est.)
3rd	AC #4 — **flagged as spiking**; model TBD	50 A	2.5–4.5 kW (est.)
3rd	AC #1 (Live Room + Iso): AC Pro PKHP-M-0601BB, 5-ton heat pump, 1-phase 230 V	50 A, 3-pole	5.5–6.5 kW (est.)
Last resort	Live-room lighting grid: LED wall + stage rig	LED wall 7.8 kW max (est.); stage rig 2–4 kW max (est.)	
?	Other interior lighting, IT, security	2–3 kW (est.); sheddable? TBD	
?	Water heater (electric? unknown)	If electric, about 4.5 kW (est.). Confirm on site	

7 ACs total ~22 kW (IES); bottom-up 24–33 kW (est.). Stagger smallest first, largest last (Greg); order is provisional. **Phase conflicts** (breaker poles vs. nameplates; staff accounts differ): **read every nameplate on site (phase, MCA/MOCP).** Sauna + EV + 7 ACs + lighting grid (last resort) = 10 controlled loads. Circuit map TBD - confirm on site.

NEVER SHED: audio, recording, network, lighting PCs. Live-room lighting grid sheds only as the very last resort.

SERVICE & UTILITY

- **Panel / service: sources conflict, CONFIRM ON SITE** (decides the controller model). IES: 600 A, 3-phase, 240 V delta, no neutral. 2024 inspection: 120/240 V, 200/400 A. Gear is **Zinsco**. Which panel feeds which meter: CONFIRM ON SITE.
- **Controlled meter 255000-006336** (SCE): rate **TOU-GS-2-E**. Bills Feb–Aug 2026 show monthly peaks of 14–21 kW.
- **Interval data, Jan 2025–Jan 2026:** peak **27.2 kW**; 50 intervals at 20.8 kW or more. Main drivers believed to be the rooftop ACs stacking (e.g. Aug 22–23, 2025) and the sauna. One 20.8 kW spike on Jul 28, 2026: cause unknown.
- **Highest combined reading, both meters:** 36.8 kW.
- Second meter: separate SCE account, max 14.4 kW, already on the cheaper rate (TOU-GS-1-E); not part of this project. Do not combine the meters.
- **Goal:** 12 straight months billed at 20 kW or less so SCE moves the meter to **TOU-GS-1-E**.

DESIRED SOLUTION

- A **demand controller** measuring the meter in real time (CTs on the customer side; nothing in the SCE metering compartment) that **sheds selected loads only**. A device that trips the main or the whole service is **not acceptable**.
- Sheds in the priority order in the table (sauna first, then EV charger, then A/C staggering), rotates, restores when there is room. Compressor minimum on/off times to prevent short-cycling; AC compressor and blower paused together.
- Controller must stay powered at all times (locked breaker); explain fail-safe behaviour if it loses power or comms.
- Sauna: confirm its panel resumes heating after a short power pause; if not, control its heater contactor instead.

CONTRACTORS

- **Blue Moon Electrical**, 310-893-5107. Does not do demand controllers.
- **High Voltage Electrical**, 310-388-4064. No documented demand controller or limiter installs.
- **CD Ampers**, 213-329-0130.

READ ON SITE

- All 7 AC nameplates: model, phase, RLA, MCA, MOCP, breaker, panel, meter
- Sauna: kW and phase
- Tesla: model
- LED wall: PSU count and breakers
- Both meter faces
- Main breaker size
- All panel directories
- Water heater: gas or electric?

ALREADY QUOTED

Existing equipment-only proposal: IES / Energy Sentry (price on request).
Oct 12, 2026: 9388BP controller (8 control points: 7 ACs + sauna, no EV), PLC transmitter + 8 receivers, 3× 400 A CTs, WattNode. Install, permit, shipping not included. Alternatives welcome.

Closest comparable job: 400 A panel with load management for 8 EV chargers (High Voltage Electrical).

OPTIONS TO QUOTE

a. **IES / Energy Sentry quote gap:** it does **not** control the Tesla Wall Connector (EV charger). How would you include the EV charger?
b. **Simpler route:** a demand controller reading CTs directly, with hardwired relays on the 7 A/C thermostat Y (cooling) wires, the sauna and the EV charger, shedding at **18.5 kW**. Please quote this too.
c. **Permit:** what permit is needed (City of Santa Monica)?
d. **License:** your CA C-10 license number.

QUESTIONS FOR YOU

1. **All-in installed price** (equipment, labor, City of Santa Monica permit, any HVAC tie-in), in writing and itemized.
2. **Timeline:** earliest start, install duration, permit lead time. Install during closed hours.
3. **How you guarantee the limit** (true 15-min average ≤19.2 kW, setpoint ~18.5 kW): response time; what if never-shed loads alone near the limit?
4. **References:** a customer with a demand controller you installed that sheds rooftop ACs.
5. Proof of insurance, warranty, and monitoring/alerts you provide.

Prepared Oct 14, 2026. TBD = unknown, confirm on site; (est.) = estimate, not measured. Sources: SCE 15-min interval data (Jan 2025–Jan 2026), IES quote #2026-008, TRC load inventory.